无人系统设计 课程作业02

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1 (4,1)起点

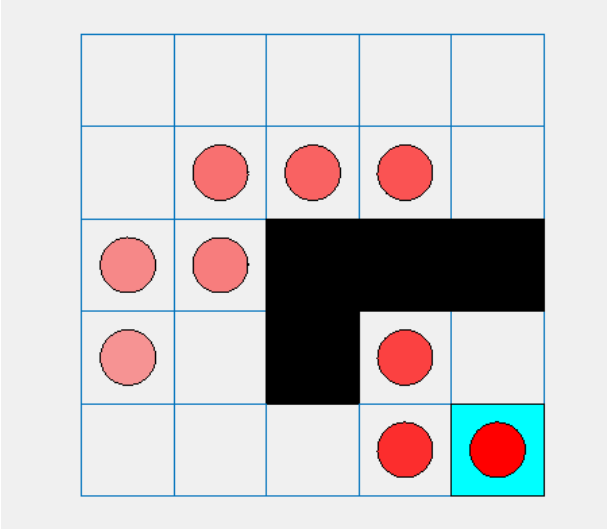
1.1 Q-Learning算法

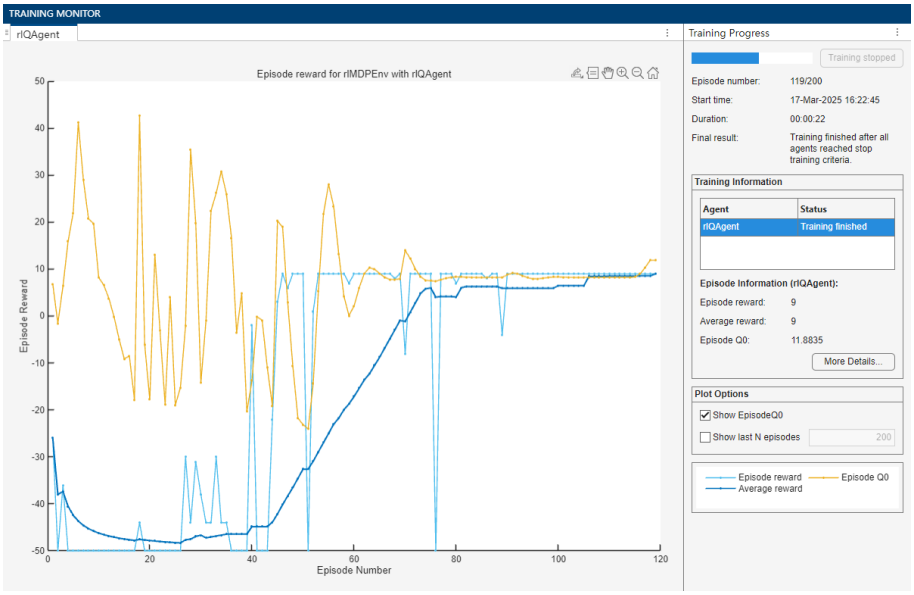
代码参数如下，经调试改变了贪婪因数和学习率。

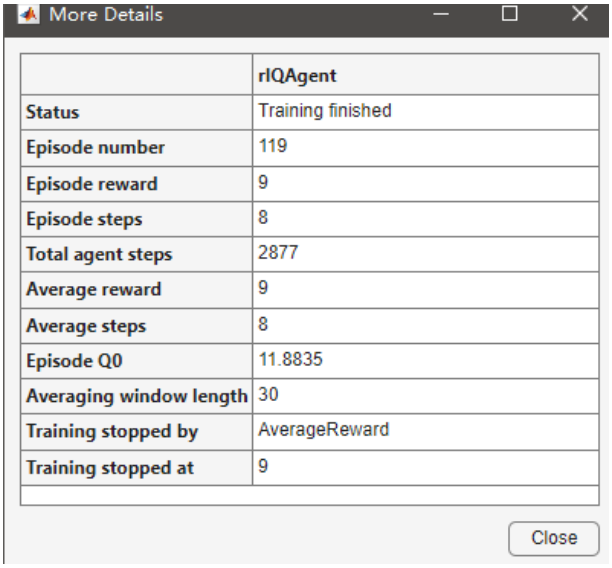
```
qAgent.AgentOptions.EpsilonGreedyExploration.Epsilon = 0.4;
qAgent.AgentOptions.CriticOptimizerOptions.LearnRate = 3;

trainOpts = rlTrainingOptions;
trainOpts.MaxStepsPerEpisode = 50;
trainOpts.MaxEpisodes= 200;
trainOpts.StopTrainingCriteria = "AverageReward";
trainOpts.StopTrainingValue = 9;
trainOpts.ScoreAveragingWindowLength = 30;
```

训练结果图如下，虽然和示例路径不一样，但是最终分数是一样的9分。







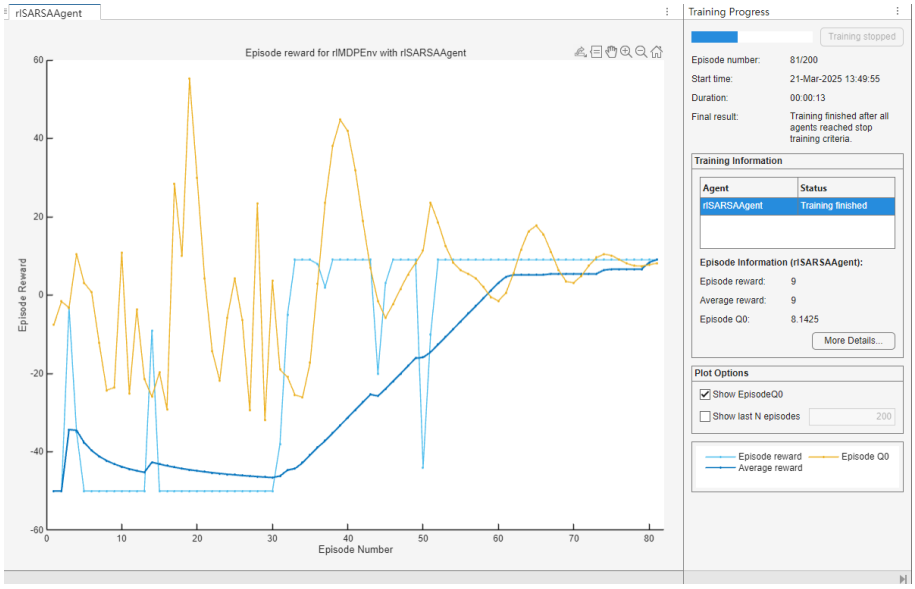
1.2 SARSA算法

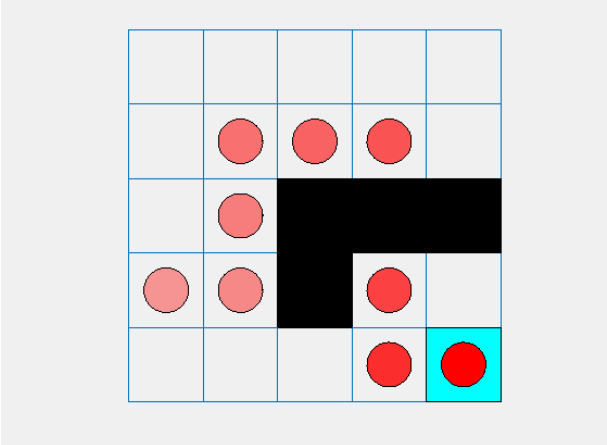
代码参数如下，调整了学习率和贪婪因子。

```
trainOpts = rlTrainingOptions;
trainOpts.MaxStepsPerEpisode = 50;
trainOpts.MaxEpisodes= 200;
trainOpts.StopTrainingCriteria = "AverageReward";
trainOpts.StopTrainingValue = 9;
trainOpts.ScoreAveragingWindowLength = 30;

sarsaAgent = rlSARSAAgent(qFcnAppx);
sarsaAgent.AgentOptions.EpsilonGreedyExploration.Epsilon = .45;
sarsaAgent.AgentOptions.CriticOptimizerOptions.LearnRate = 2.5;
```

训练结果图如下：





	rlSARSAAgent
Status	Training finished
Episode number	81
Episode reward	9
Episode steps	8
Total agent steps	2010
Average reward	9
Average steps	8
Episode Q0	8.1425
Averaging window length	30
Training stopped by	AverageReward
Training stopped at	9

Close

1.3 参数意义

- 学习率：直接影响模型更新的快慢。如果学习率太大，模型会迅速变动，可能无法找到稳定的最优路径；如果学习率太小，学习过程可能非常缓慢，甚至停滞不前。

- 贪婪因子：控制模型探索未知路径的程度。在探索阶段，模型可能会经历更多的随机选择，找到新的有效路径；在利用阶段，模型则更依赖于已有的知识，沿着最优路径前进。
- 折扣因子：影响模型在选择路径时对未来回报的关注程度。如果接近1，模型会倾向于选择那些需要时间、但长期回报更高的路径；如果接近0，模型则会选择立即获得回报的路径，可能忽略未来的潜在奖励。

1.4 心得体会

由于agent需要探索出一条距离较远的路径，所以适当调大贪婪因子与学习率，最终得到合适的参数。

2 (5,1)起点

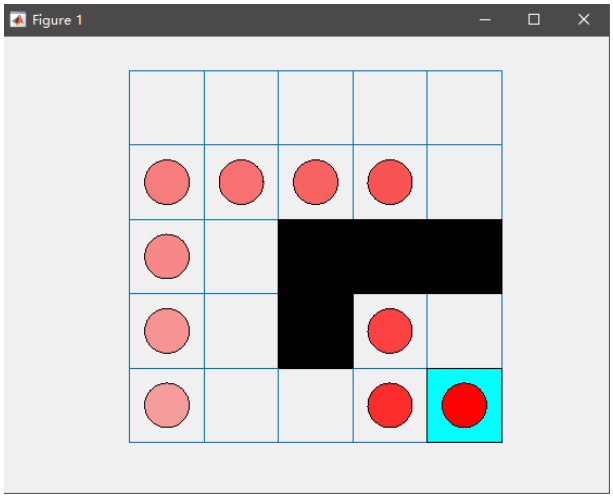
2.1 Q-Learning算法

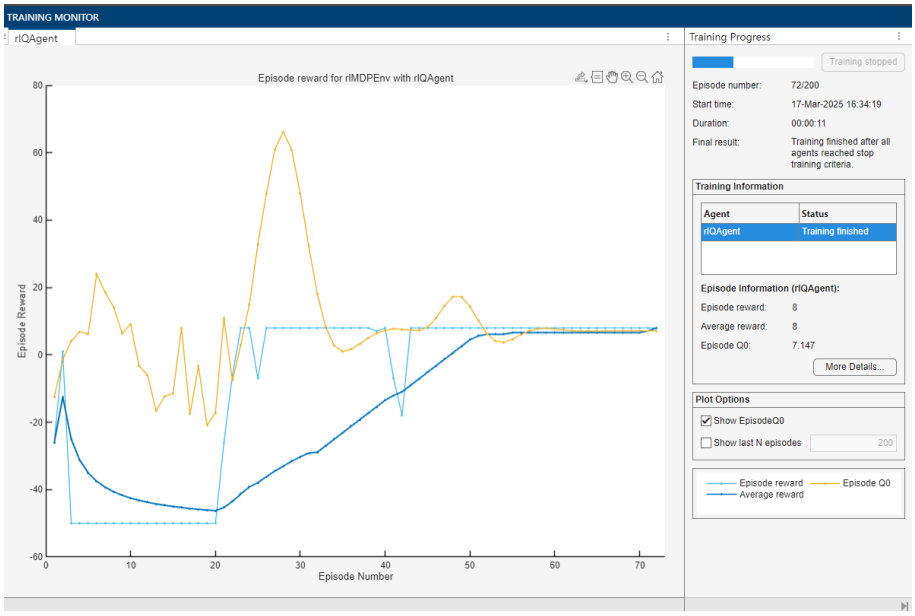
代码参数如下，经调试改变了贪婪因数和学习率。

```
qAgent.AgentOptions.EpsilonGreedyExploration.Epsilon = 0.4;
qAgent.AgentOptions.CriticOptimizerOptions.LearnRate = 2.5;

trainOpts = rlTrainingOptions;
trainOpts.MaxStepsPerEpisode = 50;
trainOpts.MaxEpisodes= 200;
trainOpts.StopTrainingCriteria = "AverageReward";
trainOpts.StopTrainingValue = 8;
trainOpts.ScoreAveragingWindowLength = 30;
```

训练结果图如下，虽然和示例路径不一样，但是最终分数是一样的8分。





	rlQAgent
Status	Training finished
Episode number	72
Episode reward	8
Episode steps	9
Total agent steps	1544
Average reward	8
Average steps	9
Episode Q0	7.147
Averaging window length	30
Training stopped by	AverageReward
Training stopped at	8

Close

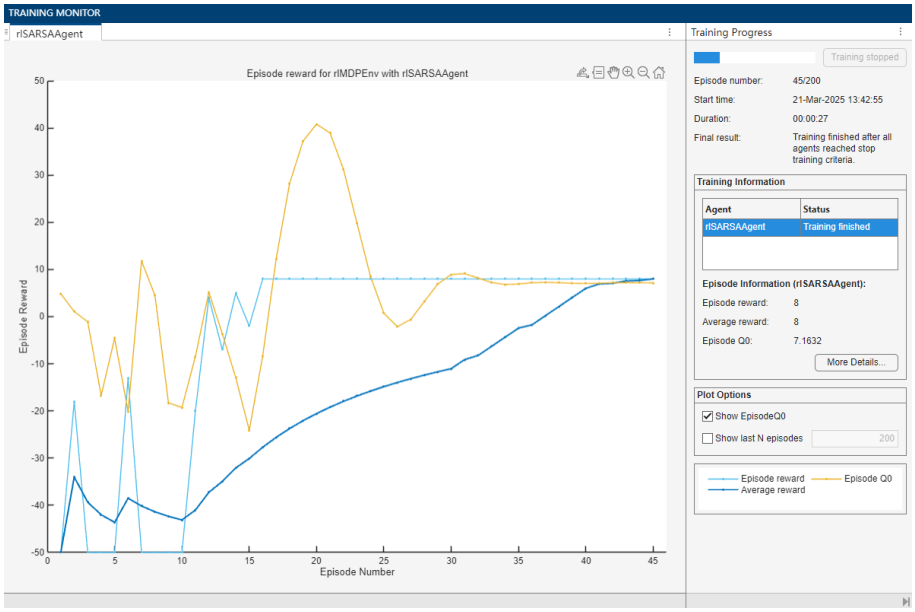
2.2 SARSA算法

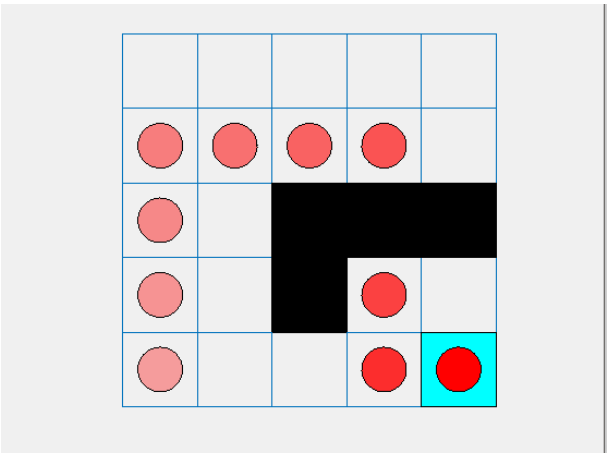
代码参数如下：

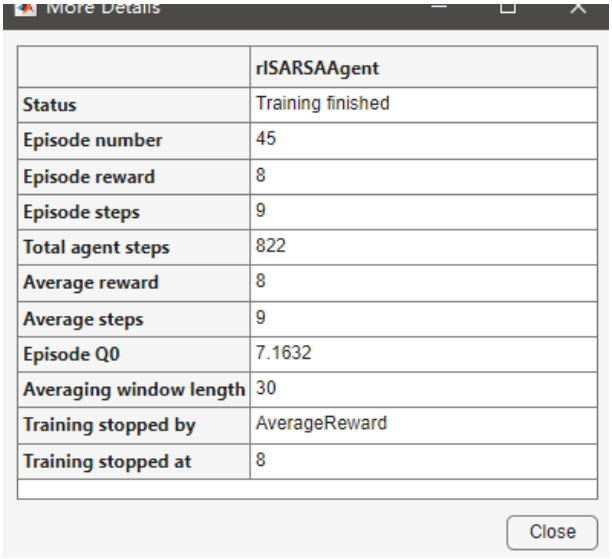
```
trainOpts = rlTrainingOptions;
trainOpts.MaxStepsPerEpisode = 50;
trainOpts.MaxEpisodes= 200;
trainOpts.StopTrainingCriteria = "AverageReward";
trainOpts.StopTrainingValue = 8;
trainOpts.ScoreAveragingWindowLength = 30;

sarsaAgent = rlSARSAAgent(qFcnAppx);
sarsaAgent.AgentOptions.EpsilonGreedyExploration.Epsilon = .45;
sarsaAgent.AgentOptions.CriticOptimizerOptions.LearnRate = 2.5;
```

训练结果图如下：







2.3 参数意义

见第一部分。

2.4 心得体会

与第一部分类似，由于5,1坐标离最优路径更远了，需要更大的调整才能找到合适的参数。